ITM University Gwalior

SchoolofComputingandIntelligentSystems

Project-Based Learning (PBL)

Mini Project Report

Student Interests and Engineering Branch Preferences Analysis

Course: DataAnalytics Using Python

Course Code: CSL0312

Program: B.Tech.CSE (Artificial Intelligence & Machine Learning)

Semester: III Section-F

Academic Session: July–Dec 2026

Project Details

Project Title: StudentInterestsandEngineeringBranchPreferencesAnalysis

Domain (as per Section 6): Education / Student Preference Analytics

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Date of Submission:

1. Introduction & Problem Statement

Choosing an engineering branch is oneof the most important academic decisions a student makes, and it shapes later career options. The choice is usually influenced by several factors at once: favourite subjects, technical skills, hobbies, career goals, family advice, salary expectations and industry demand. When educators and academic advisors understand which preferences are common and how they relate to each other, they can plan better counselling, orientation and mentoring activities.

This project analyses a student-preference dataset using Python (Pandas, NumPy, Matplotlib and Seaborn). The data comes from two files that were loaded in the notebook: an academic profile file (7 columns) and a career preference file (10 columns). Both contain 2,000 students and were merged on Student_ID into a single table of 2,000 records. The raw data contained inconsistent text formatting (for example “KARNATAKA”, “M”, “F”, “yrs”, “%”) and 980 missing cells, which had to be cleaned before analysis.

The problem statement is: “Which engineering branches, interest areas, skills, career goals and influencing factors are most common among the students in the dataset, and how are interests associated with the branches they prefer?” The scope is descriptive and exploratory. It covers data cleaning, frequency analysis, cross-tabulation, group-wise comparison, visualisation and basic statistics. No predictive model is built, and no causal claim is made. The records are labelled Synthetic/Simulated in the Data_Source_Type column, so the findings describe patterns in this dataset and should not be generalised to all students.

2. Dataset Description & Source

Dataset Name: StudentInterestsandEngineering Branch Preferences (academic_profiles + career_preferences)

Source / Organization: Data_Source_Type = “Synthetic/Simulated”

Dataset Link (URL):

Typeof Dataset: □ Public □ Private □ Self-Collected (the data is recorded as Synthetic/Simulated)

Number ofRecords: 2,000

Number ofAttributes/Columns: 16 in the merged table (15 distinct fields; Data_Source_Type appears once from each source file)

☒ **Dataset meets minimum quality criteria (Section 7):** 2,000 records (minimum 500), 16 columns (minimum 5). Missing values are low (980 of 32,000 cells, about 3.06%) and are documented and treated in Section 3.

Brief description of attributes/columns

Attribute	Type	Description
Student_ID	Identifier	Unique ID (STU0001 …); join key between the two files
Age	Numeric	Age in years (stored as text in raw data; some values carried “yrs”)
State	Categorical	State of the student (e.g. Karnataka, Maharashtra, Delhi) Gender
Gender	Categorical	as recorded (Male, Female, Other, Prefer Not To Say) Class 12
Class_12_Percentage	Numeric	marks in percent (raw values could include “%”) Subject the
Favourite_Subject	Categorical	student reports as favourite
Interest_Area	Categorical	Main area of interest (e.g. Software Development, Electronics)
Technical_Skill	Categorical	Main technical skill (e.g. Programming, Python, CAD Design)
Hobby	Categorical	Reported hobby (e.g. Coding, Robotics, Reading)

Attribute	Type	Description
Preferred_Engineering_Branch	Categorical	Branch the student prefers (16 categories plus Unknown)
Career_Goal	Categorical	Intended career (e.g. Researcher, Entrepreneur)
Main_Choice_Influence	Categorical	Main factor behind the branch choice
Preferred_Work_Area	Categorical	Preferred sector (e.g. Software Industry, Research, Startup)
Reason_for_Branch_Choice	Categorical	Stated reason for the choice (e.g. Passion for the field)
Data_Source_Type (×2)	Categorical	Source label, "Synthetic/Simulated"; kept as _Academic and _Career after merge

3. Data Cleaning & Preprocessing

The two files were first inspected separately (shape, column names, data types, missing values, duplicates and Student_ID uniqueness) and then merged with an inner join. A working copy (df_clean) was created so that the raw data stays unchanged. The steps are summarised below.

3.1 Initial quality checks

- Academic file: 2,000 × 7 with 340 missing cells. Career file: 2,000 × 10 with 640 missing cells.
- No duplicate rows in either file, and all 2,000 Student_ID values were unique in both files.
- Inner join on Student_ID kept all 2,000 records, giving a merged table of 2,000 × 16 (no students lost).
- Age, Class_12_Percentage and other numeric fields were loaded as text (object type), so type conversion was needed.

3.2 Cleaning steps

- **Whitespace:** Leading and trailing spaces were removed from all text columns.
- **Case standardisation:** Title case was applied to 11 categorical columns so that "KARNATAKA", "karnataka" and "Karnataka" become one category.
- **Label harmonisation:** Gender values "M" and "F" were mapped to Male and Female, and "Not Specified" to Prefer Not To Say.
- **Numeric conversion:** The text "yrs" was removed from Age and "%" from Class_12_Percentage; both columns were then converted with pd.to_numeric(errors="coerce").
- **Missing values:** Missing Age and Class_12_Percentage values were filled with the column median; missing categorical values were filled with "Unknown". No missing values remained afterwards.
- **Branch aliases:** Preferred_Engineering_Branch labels "Ai And Ml" (210) and "Ai & Ml" (115) were merged into "AI and ML" (325), and capitalisation of "Computer Science and Engineering" was fixed.
- **Outlier / range check:** Ages outside 15–30 and percentages outside 0–100 were checked; none were found.

3.3 Missing values before cleaning

Column(s)	Missing count	Missing %
Class_12_Percentage	100	5.0%
Gender	80	4.0%

Column(s)	Missing count	Missing %
Interest_Area, Technical_Skill, Hobby, Preferred_Engineering_Branch, Career_Goal, Main_Choice_Influence, Preferred_Work_Area, Reason_for_Branch_Choice (each)	80	4.0%
Age, Favourite_Subject (each)	60	3.0%
State	40	2.0%
Student_ID, both Data_Source_Type columns	0	0.0%

3.4 Representative code

```
df = pd.merge(academic_df, career_df, on="Student_ID", how="inner",
              suffixes=("_Academic", "_Career"))
df_clean = df.copy()

for col in df_clean.select_dtypes(include="object").columns:
    df_clean[col] = df_clean[col].str.strip()
for col in category_columns:
    df_clean[col] = df_clean[col].str.title()

df_clean["Gender"] = df_clean["Gender"].replace(
    {"M": "Male", "F": "Female", "Not Specified": "Prefer Not To Say"})
df_clean["Age"] = pd.to_numeric(
    df_clean["Age"].astype("string").str.replace("yrs", "").str.strip(),
    errors="coerce")
df_clean["Class_12_Percentage"] = pd.to_numeric(
    df_clean["Class_12_Percentage"].astype("string")
    .str.replace("%", "").str.strip(), errors="coerce")

for col in ["Age", "Class_12_Percentage"]:
    df_clean[col] = df_clean[col].fillna(df_clean[col].median())
for col in df_clean.select_dtypes(include="object").columns:
    df_clean[col] = df_clean[col].fillna("Unknown")

df_clean.to_csv("cleaned_student_preferences.csv", index=False)
```

3.5 Result and limitations

Aftercleaning, the dataset has 2,000 rows and 16 columns, 0 missing values, 0 duplicate rows and 0 duplicate Student_ID values. Two limitations should be kept in mind. First, “Unknown” is a real category in the counts (80 records for most career fields, and 80 for Gender), so it is excluded from the charts but not from the data. Second, median imputation puts 100 students at the same percentage (77.95) and 60 at the same age (19), which slightly reduces the spread of these variables. A small number of Gender entries recorded as “Pns” (8) were not merged into “Prefer Not To Say” (29) and remain a separate label.

4. Exploratory Data Analysis (EDA)

EDA was carried out on the cleaned data using frequency counts and percentages for the main categorical fields, summary statistics for the two numeric fields, a cross-tabulation of Interest_Area against Preferred_Engineering_Branch, and group-wise Class 12 statistics by branch.

4.1 Preferred engineering branch

Branch (top 8)	Students	Share
Computer Science and Engineering	483	24.15%
AI and ML	325	16.25%
Information Technology	210	10.50%
Data Science	199	9.95%
Mechanical Engineering	102	5.10%
Electronics And Communication Engineering	100	5.00%
Electrical And Electronics Engineering	90	4.50%
Cybersecurity / Civil Engineering (each)	65	3.25%

Computer-related branches dominate. CSE, AI and ML, IT and Data Science together account for 1,217 of 2,000 students (60.85%). Before the "AI and ML" aliases were merged, the largest label was CSE with 483 and the two AI labels had 210 and 115; after merging, AI and ML rises to second place with 325.

4.2 Interest areas, skills, career goals and influences

Interest area	Students	Technical skill	Students
Software Development	384 (19.20%)	Programming	289 (14.45%)
Artificial Intelligence	307 (15.35%)	Data Analysis	228 (11.40%)
Data Analysis	298 (14.90%)	Python	221 (11.05%)
Electronics	272 (13.60%)	Web Development	130 (6.50%)
Cybersecurity	202 (10.10%)	Problem Solving	129 (6.45%)

Career goal	Students	Main choice influence	Students
Researcher	455 (22.75%)	Personal Interest	585 (29.25%)
Entrepreneur	306 (15.30%)	Career Opportunities	417 (20.85%)
Software Developer	289 (14.45%)	Subject Strength	220 (11.00%)
Data Scientist	202 (10.10%)	Family Advice	219 (10.95%)
App Developer	106 (5.30%)	Salary Expectations	216 (10.80%)

Software Development is the leading interest area, and Programming, Data Analysis and Python are the leading skills. Researcher is the most common career goal, ahead of Entrepreneur and Software Developer. Personal Interest (29.25%) is the most common influence, followed by Career Opportunities (20.85%). Peer Influence is the lowest recorded factor (116 students, 5.80%).

4.3 Interest area versus preferred branch (cross-tabulation)

The cross-tabulation shows that each interest area is linked to a small set of related branches rather than to all branches:

- Software Development → IT (135), AI and ML (117), CSE (113).
- Artificial Intelligence → CSE (108), AI and ML (99), Data Science (92).

- Data Analysis → Data Science (99), AI and ML (97), CSE (93).
- Electronics → ECE (96), EEE (89), CSE (75).
- Cybersecurity → CSE (66), IT (65), Cybersecurity (62).
- Mechanical Design → Mechanical (62), Mechatronics (60), Automobile (57). •

Civil Infrastructure → Civil (64), Environmental (46), Mechanical (38).

- Biotechnology → Biotechnology (39), Chemical (38), Biomedical (35).

The students whose interest area is Unknown (80) are spread across branches, with CSE the largest share (28). This very clean grouping suggests that the simulated data was generated with rules linking interests to branches; real survey data would normally show more mixing.

4.4 Class 12 percentage by preferred branch

Excluding Unknown, average Class 12 percentage ranges from 76.09 (Biotechnology, n = 39) to 80.35 (Civil Engineering, n = 65). The two largest groups, CSE (n = 483, mean 78.81) and AI and ML (n = 325, mean 77.77), are close to the overall mean of 78.16. Because the overall standard deviation is about 8.70, these differences between branch means are small.

5. Data Visualization & Observations

The charts below were generated in the notebook with Matplotlib and Seaborn from the cleaned data. “Unknown” was excluded from the category charts.

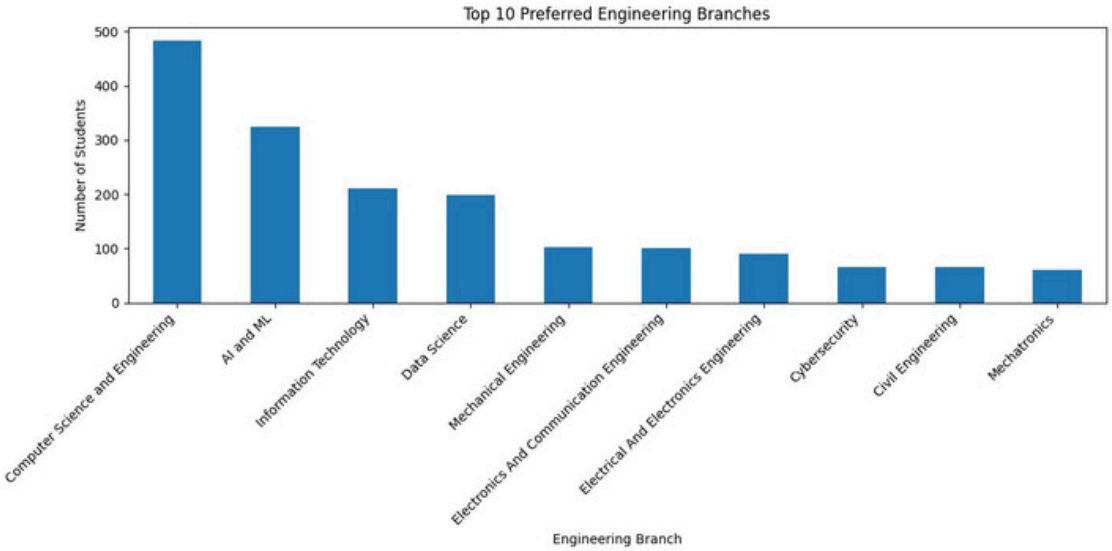


Figure 1.Top10preferredengineeringbranches.

Observation: CSE is the clear leader, followed by AI and ML, Information Technology and Data Science. Core branches such as Mechanical, ECE and EEE come next with roughly 90–100 students each.

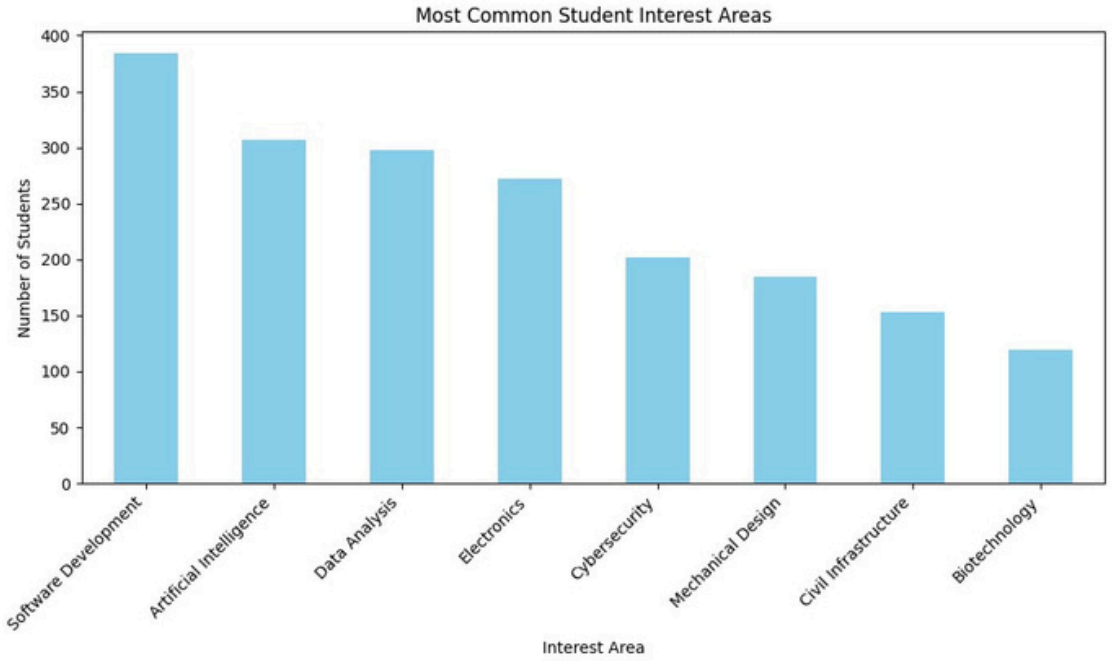


Figure 2. Most commonstudent interest areas.

Observation: Software Development, Artificial Intelligence and Data Analysis are the top three interests, and together they cover about 49% of all students. Biotechnology and Civil Infrastructure are the least common.



Figure 3. Heatmap of interest areaversus preferred engineering branch.

Observation: Dark cells show the most common combinations: Software Development with IT, AI and ML and CSE, and Electronics with ECE and EEE. Most cells are empty, so each interest maps to a few related branches. This shows association in the dataset, not that an interest causes a branch choice.

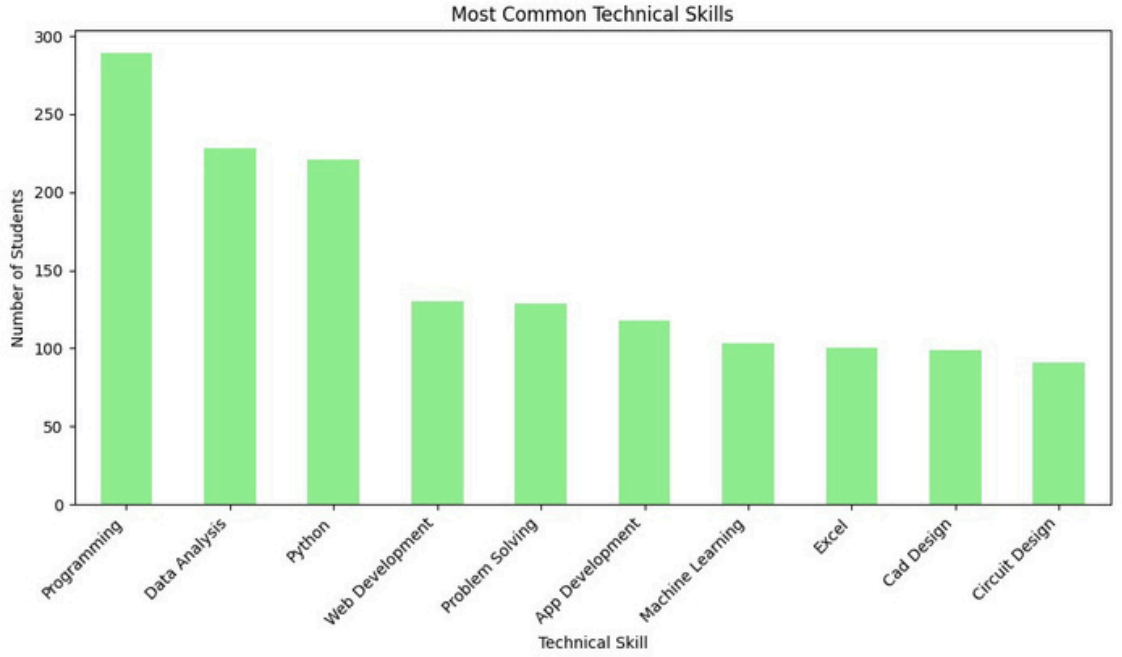


Figure 4. Most common technical skills.

Observation: Programming, Data Analysis and Python are the three most common skills. Skills linkedto core branches (CAD Design, Circuit Design, Embedded Systems) appear at about 88–99 students each.

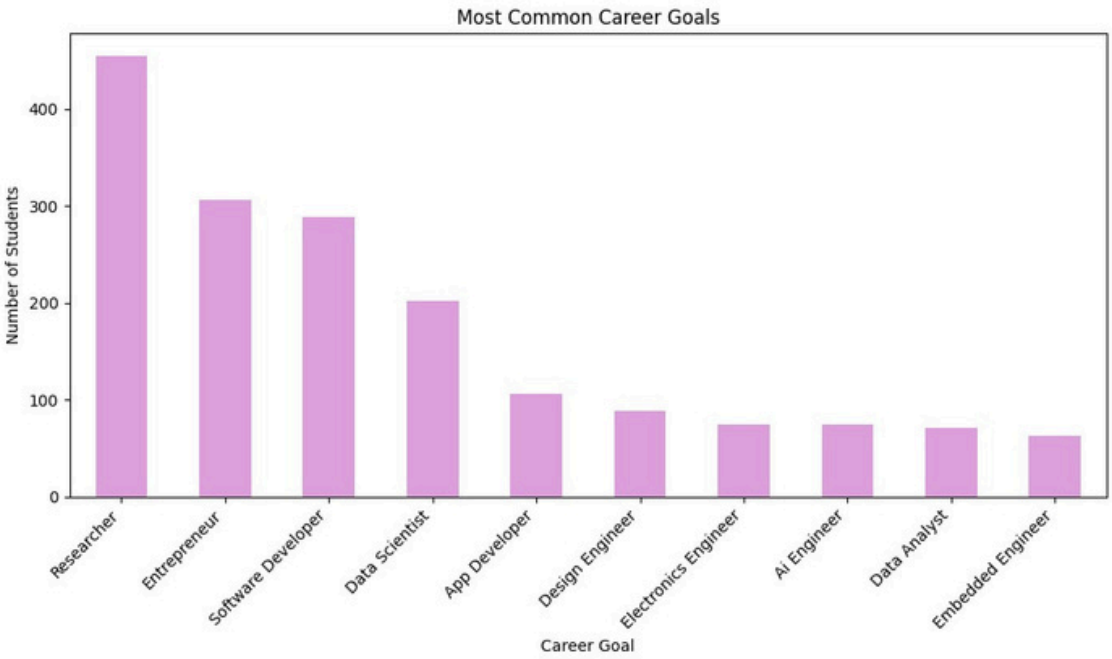


Figure 5. Most common career goals.

Observation: Researcher is the most frequent goal (455 students), well ahead of Entrepreneur (306) and Software Developer (289). Government Service and Biotechnology Researcher are the least common.

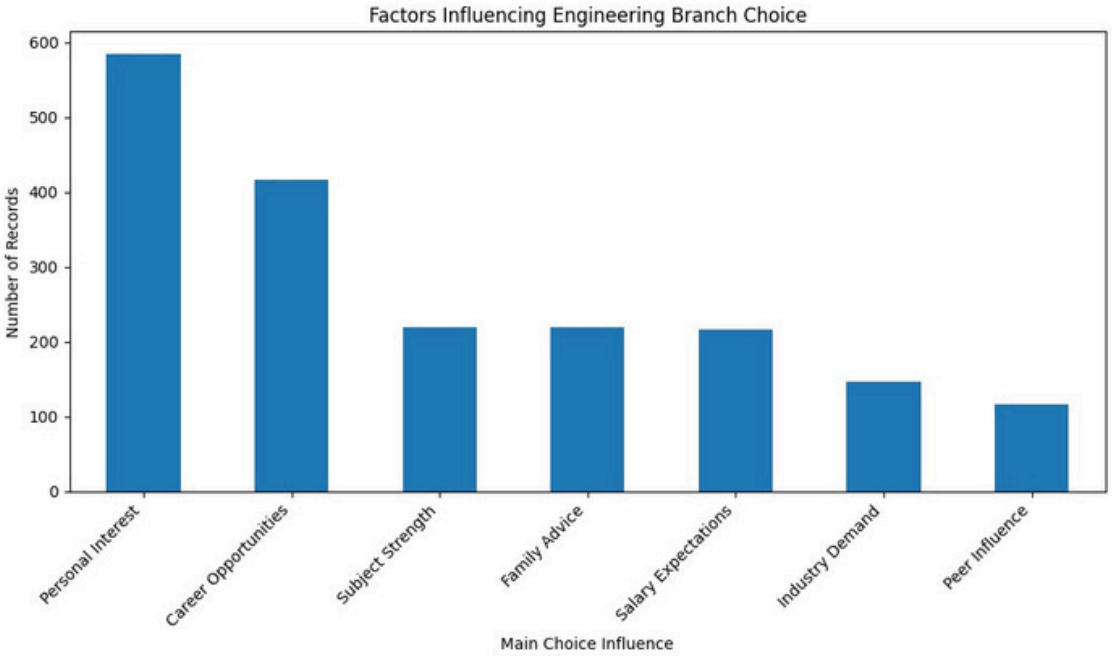


Figure 6. Factors influencing engineering branch choice.

Observation: Personal Interest is the main influence for 585 records, followed by Career Opportunities(417). Subject Strength, Family Advice and Salary Expectations are similar in size at about 216–220 each.

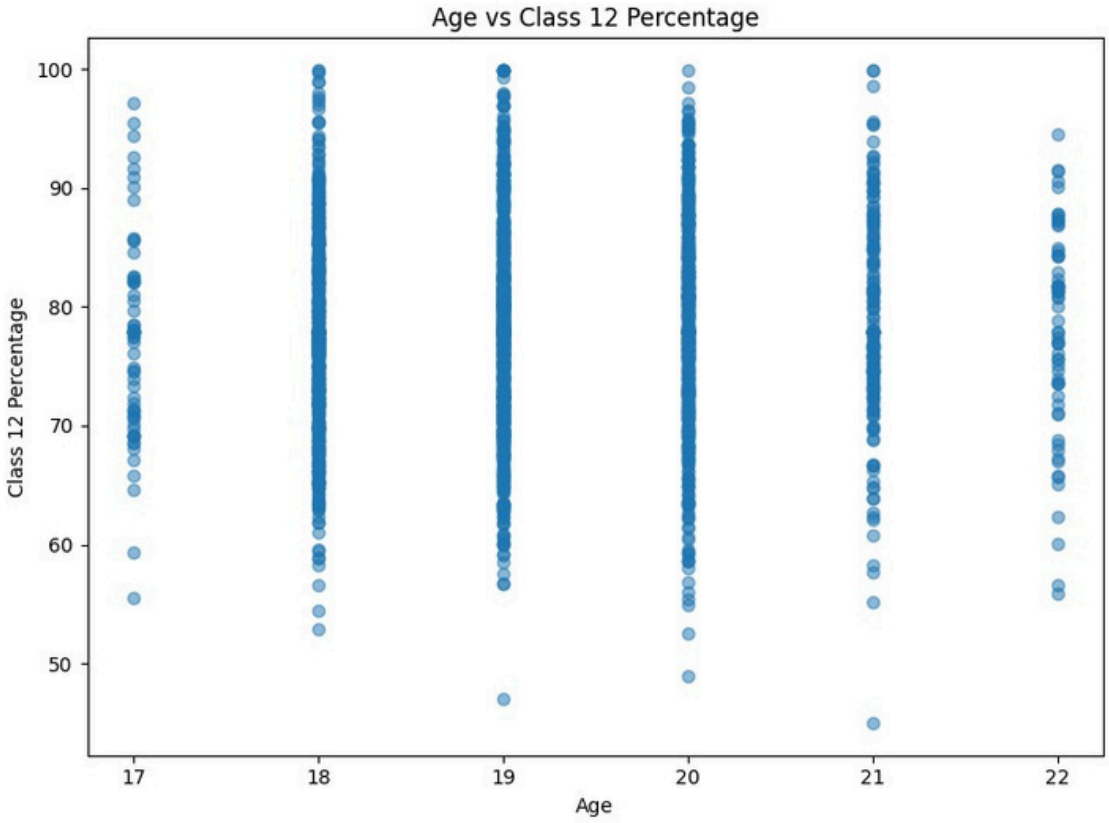


Figure 7. Scatter plot of age versus Class 12 percentage.

Observation: The points are spread widely across all ages (17–22) and percentages, with no visibleupward or downward trend. Age takes whole-number values, so the points form vertical columns. This agreeswith the very low correlation reported in Section 6.

6. Statistical Analysis (Basic)

6.1 Descriptive statistics

Statistic	Age (years)	Class 12 percentage
Count	2,000	2,000
Mean	19.20	78.16
Standard deviation	1.09	8.70
Minimum	17	45.00
25th percentile (Q1)	18	72.37
Median	19	77.95
75th percentile (Q3)	20	83.94
Maximum	22	99.90

The average student is about 19.2 years old, and the middle 50% are between 18 and 20. The mean Class 12 percentage (78.16) is close to the median (77.95), which suggests a fairly symmetric distribution. The middle 50% of students scored between 72.37 and 83.94, with a standard deviation of about 8.7 percentage points.

6.2 Correlation

The Pearson correlation between Age and Class_12_Percentage is r = 0.031, which is very close to zero. Using n = 2,000, this gives t ≈ 1.39 and a two-sided p-value of about 0.17 (computed from the reported r and n). At the 5% level this is not statistically significant, so the data gives no evidence of a linear relationship between age and Class 12 marks. Part of the data (60 ages and 100 percentages) was median-imputed, which can pull the correlation slightly towards zero.

6.3 Group-wise comparison

Mean Class 12 percentage was compared across preferred branches (Section 4.4). Means lie between 76.09 and 80.35, a range of about 4.3 points against a standard deviation of 8.7, and the smallest groups have only 37–65 students. No hypothesis test was applied to these group differences, so they are described only and should not be read as significant.

7. Key Insights & Interpretation

1. Computer-related branches dominate preferences. CSE (24.15%), AI and ML (16.25%), IT (10.50%) and Data Science (9.95%) together cover about 61% of students, which shows strong demand for computing and data fields in this dataset.

2. Merging branch aliases changes the ranking. Combining “Ai And Ml” and “Ai & Ml” moved AI and ML from 210 to 325 and into second place, so consistent labels are important before comparing categories.

3. Interests map to related branches. Each interest area leads to only two or three branches (for example Mechanical Design to Mechanical, Mechatronics and Automobile), so interest is a useful guide for counselling in this dataset. Because the data is simulated, this tidy pattern may be more rigid than real life.

4. Personal interest and career opportunities lead the influences. Personal Interest (29.25%) and Career Opportunities (20.85%) together cover about half of the students, while Peer Influence is the smallest at 5.80%. Guidance programmes could therefore focus on helping students match their interests to career prospects.

5. Class 12 marks and age show no meaningful link. The correlation of 0.031 is not significant (p ≈ 0.17), and average marks differ only slightly between branches (76.09 to 80.35), so marks alone do not explain branch preference here.

8. Conclusion & Future Scope

This project built a complete Python workflow for a student-preference dataset: merging two files of 2,000 students, cleaning 980 missing cells and inconsistent text and number formats, merging branch aliases, exploring frequencies and cross-tabulations, drawing seven charts, and computing basic statistics. The analysis shows that CSE, AI and ML, IT and Data Science are the most preferred branches, Software Development is the top interest, Researcher is the top career goal, and Personal Interest is the main influence on choice. Class 12 percentage and age show no meaningful relationship. Since the dataset is labelled Synthetic/Simulated, these findings should not be generalised to the wider student population without validation on real data.

Future scope: (1) consolidate the remaining category aliases (for example “Pns” in Gender) with an explicit mapping dictionary; (2) document how the dataset was generated and add its source link; (3) analyse why values are missing and compare imputation methods; (4) add chi-square tests for interest versus branch and

ANOVA for marks across branches; (5) build an interactive dashboard for counsellors; and (6) if prediction is wanted, define a target variable, split the data into training and test sets, and evaluate a classification model.

9. References

1.PBL_notebook (Python analysis notebook, Google Colab) — datasets academic_profiles and career_preferences, accessed October 2026.

2. ITM University Gwalior, School of Computing and Intelligent Systems. PBL Mini Project Report — Submission Format, Course CSL0312, Session July–December 2026.

3. pandas development team. pandas documentation. https://pandas.pydata.org/docs/

4. Hunter, J. D. Matplotlib documentation. https://matplotlib.org/stable/

5. Waskom, M. seaborn documentation. https://seaborn.pydata.org/

6. NumPy documentation. https://numpy.org/doc/

10. AI Tool Usage Declaration

AsperSection4&15ofthePBLguidelines,any use of AI tools must be transparently declared below.

□ No AI tools were used in this project.

☒ AI tools were used, as detailed below:

AI Tool Used	Purpose	Extent of Use / Section Applied
Chat gpt	Coding guidance,debugging	Colab workflow, explanation,report organization and documentation

11. Project Links & Deliverables

GitHub RepositoryLink: https://github.com/Mahesh204505/student-preference-datasets
Python Notebook (.ipynb) File Name: PBL_notebook.ipynb
Cleaned Dataset File: cleaned_student_preferences.csv
PBL Report File Name: PBL_Report.docx

□ GitHub repository includes code, notebook, PBL report and README (tick after uploading).

12. Report Structure Checklist (Section 8 Compliance)

S. No.	Report Section (as per Section 8 format)	Included (✓)
1	Title Page	✓
2	Introduction & Problem Statement	✓

S. No.	Report Section (as per Section 8 format)	Included (✓)
3	Dataset Description & Source	✓
4	Data Cleaning & Preprocessing	✓
5	Exploratory Data Analysis (EDA)	✓
6	Data Visualization & Observations	✓
7	Statistical Analysis (Basic)	✓
8	Key Insights & Interpretation	✓
9	Conclusion & Future Scope	✓
10	References	✓

13. Declaration by Student(s)

I/We hereby declare that thisMini Project Report is our own original work, prepared using publicly available/open datasets. All sources of data, references, and AI tool assistance have been duly acknowledged. I/We understand that any instance of plagiarism in code or report will attract strict academic action, as per Section 15 of the PBL guidelines.

Team Leader Signature: _____

Team Member Signature(s): _____

Date: _____

14. For Faculty Use Only — PBL Final Report Evaluation

EvaluationComponentB—PBLFinalReportSubmission | Mid-Term(PBLBased):10 Marks (Section 10 of PBL Guidelines)

Evaluation Criteria	Max Marks	Marks Awarded
Data Cleaning & Preprocessing	2	
Exploratory Data Analysis (EDA)	2	
Visualization & Observations	2	
Statistical Analysis	2	
Insights, Interpretation & Conclusion	2	
Total (out of 10)	10	

Faculty Remarks:

Faculty Mentor Signature: _____

Date: _____